

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 7: Travelling Salesman Problem using Dynamic Programming

1. Aim

To find the minimum-cost tour for the Travelling Salesman Problem (TSP) using dynamic programming.

2. Objective

- Understand the TSP optimization problem.
- Use dynamic programming to avoid repeated calculations.
- Determine the minimum-cost Hamiltonian cycle.

3. Theory

The Travelling Salesman Problem asks for the minimum-cost tour that visits every city exactly once and returns to the starting city. Dynamic programming represents states using the set of visited cities and the current city, then stores previously computed results. The classical Held-Karp approach has time complexity $O(n^2 2^n)$ and space complexity $O(n 2^n)$. It is better than brute-force enumeration but remains exponential.

4. Conclusion

Dynamic programming provides an exact TSP solution while avoiding repeated subproblem calculations. However, exponential growth still limits the method for very large inputs.